

A uniform strong-coupling gap, heat flow and complementary relaxation for the SU(2) Kogut-Susskind Hamiltonian: an audited record of the Yang-Mills workbench

Claude (Anthropic, model claude-opus-5-5), 27 September 2026. Published at the author's request ahead of the author's review; the author may revise or withdraw it. This record replaces the Yang-Mills reader (Zenodo 10.5281/zenodo.22987531) and note 46_, which were withdrawn on 27 September 2026.

Abstract

The Yang-Mills workbench of the author (KokunoYumeto/yang-mills-interacting-workbench, Zenodo record 10.5281/zenodo.22883643) studies the Kogut-Susskind Hamiltonian of SU(2) lattice gauge theory on open cubic boxes. Its central result is an explicit lower bound on the spectral gap above the ground state, on the whole gauge-invariant space, uniform in the box size and the lattice spacing, for bare coupling $g^2 \geq 3.6836$ in the workbench's normalization. We give the proof, with a sharp form of its key Casimir inequality: on any graph the best uniform constant is the girth. With inputs recomputed in this audit, the same argument gives a gap of at least 1.52κ for $g^2 \geq 3.9781$, which includes the workbench's benchmark $g^2 = 4$. We add a bound at every truncation order, the gap on the full (non-gauge-invariant) space, the persistence of the gap at complex coupling, so that the ground state is analytic on a disk of box-independent radius, and an upper bound that locates the gap in $[2.90\kappa, 3.0003\kappa]$ for $g^2 \geq 13$. For the heat flow of plaquettes we re-derive every constant of the box-independent heat remainder, improve its prefactor from 67896/169 to 2829/13, derive the kinetic-row constants, and sharpen the relaxation bounds into the complementary physical space. We also state the proved limits of the global perturbative route and of the workbench's continuum path, and the typed maps from the workbench to the Jacobian, Navier-Stokes and S^6 material.

0. Introduction

0.1 Why this record

The Clay problem asks for a quantum Yang-Mills theory on $\mathbb{R}^4$ with a mass gap [JW]. Lattice gauge theory approaches it through regulated models, and at strong coupling mass gaps are classical in the Euclidean setting [OS]. The workbench works in the Hamiltonian setting of Kogut and Susskind [KS] and aims at explicit bounds that are uniform in the volume, on the full physical space, with closed coupling thresholds, together with the machinery that such bounds make available: heat correlations, relaxation into the complementary space, and analyticity in the coupling. Its material spans five Zenodo editions and several hundred pages. This record states what is proved, with the proofs, and what it means.

0.2 Main results

- **Theorem 2.1** (the workbench's fifth-reference gap bound). $\Delta_L \geq \kappa d_5(\xi)$ for every box $L \geq 2$, every spacing $a > 0$ and every $\xi \leq \alpha_5$, that is $g^2 \geq 3.6836$; in particular $\Delta_L \geq 1.5453\kappa$ on that range and $\Delta_L > 1.9068\kappa$ for $g^2 \geq 4$. Conditional on the computed coefficient bounds of orders 3-5.
- **Lemma 2.2** (*generalized here*). On a graph of girth g , gauge invariance forces $\Sigma_f j_f \geq g j_e$ on every Peter-Weyl block, and the girth is the best constant uniform over the links; Remark 2.3 gives the exact constant for each link.
- **Corollary 2.6** (*generalized here*). The gap bound at every truncation order N ; with audited inputs of orders 1-2 it holds for $g^2 \geq 3.9781$ with $\Delta_L \geq 1.52\kappa$.
- **Corollary 2.7 and Proposition 2.8** (*new*). The gap on the full space is at least $\kappa d_N/4$, sharp at $\xi = 0$; at complex coupling $|\zeta| \leq \alpha_N$ the ground state is algebraically simple and separated by a strip, so it is analytic in the coupling on a disk whose radius does not depend on the box.
- **Theorem 3.1** (*improved here*). A box-independent heat remainder on $|\xi| < 1/55$ with prefactor 2829/13 and rate 13/8, and a family of radii.
- **Propositions 3.3-3.5**. Relaxation bounds into the complementary physical space, the derived kinetic-row constants, and the benchmark tables: at $g^2 \geq 13$ the energy loss is below 0.000237 and at $g^2 \geq 16$ below 0.0000194.

- **Proposition 3.6** (*new*). An upper bound on the gap from plaquette states, which locates the gap in $[2.9047, 3.00024]\kappa$ for $g^2 \geq 13$ and shows that the true gap has no first-order term in ξ .

0.3 What the results mean

At strong coupling the workbench obtains, for the SU(2) Hamiltonian on open boxes, a spectral gap with explicit constants that is uniform in the volume and holds on the entire physical space. The proof rests on two structural facts: gauge invariance forces the Casimir of every nonconstant physical block to be at least 3 (the girth of the cubic lattice is 4), and the vacuum, written as an exponential, can be controlled in local Fourier-algebra norms that do not grow with the box. The same control gives box-independent heat correlations, near-complete relaxation of the plaquette family, analyticity of the ground state in the coupling on a box-independent disk, and an upper bound that pins the gap within 0.03% of 3κ at $g^2 \geq 13$.

These are statements at fixed lattice spacing and strong bare coupling. The continuum problem requires a $\rightarrow 0$ with $g \rightarrow 0$, the opposite regime, and the workbench’s standing continuum path leaves the domain of these estimates after finitely many steps (Proposition 6.2). What would connect the two regimes is a bound of this kind that persists along a path to weak coupling; the questions in §8 state precisely which ingredients of the present proof would have to be extended. The workbench also pursues the opposite construction: sequences driven by the S^6 period data in which box, spacing and coupling vary together, aimed at models that are gapped at every regulator but whose continuum limit is gapless (§7).

0.4 How this record was made

This is an audit, carried out by an AI system, of research that was itself carried out with AI systems. The workbench credits its tools to the literature: the exponential form of the vacuum and its character calculus to Schütte, Zheng and Hamer [SZH], the Fourier-algebra estimates to Eymard [Ey], semigroup generation to Lumer and Phillips [LP], and the creation method of its first volume-uniform gap to Yarotsky [Ya], with Baez’s spin networks [Ba] for the physical space. Two inputs of side branches are credited by the workbench to Levent Alpöge: the Jacobian-conjecture counterexample (whose announcement credits Akhil for the question and Fable for the work) and the S^6 construction [A]. The Navier–Stokes material rests on OpenAI’s manuscript [O].

The audit read the two proof manuscripts of the latest edition in full (F1–F42 and H1–H34), re-derived the analytic arguments, recomputed the constants and thresholds in exact arithmetic, recomputed the order-2 inputs from explicit SU(2) tensors, re-ran the workbench’s order-5 producer and auditor with byte-identical output, and added the results marked *new* or *generalized here*. Statements carry one of four statuses, as in the companion papers: *verified* (proved here or re-computed by a program listed in §9), *not verified here*, *my assessment* (a view with its reasons, in the first person), and *proved negative* (with the counterexample or derivation in the text). Where a result depends on inputs that were not recomputed, it is stated as conditional on them, and the inputs are named.

0.5 Organization

Part A contains the verified results: the setting (§1), the gap (§2), heat flow and relaxation (§3), the earlier editions (§4), and the typed maps from the side branches (§5). Part B (§6) contains the proved limits of specific routes. Part C (§7) describes the directions that remain open, with my assessment. §8 lists questions and §9 the verification.

Part A. Verified results

1. The setting

Box and space. The vertices form the open box $\{-L, \dots, L\}^3$, $L \geq 2$. The links e are the positively oriented nearest-neighbour edges inside the box and the plaquettes p the elementary faces inside it; there are $M_L = 12L^2(2L + 1)$ plaquettes. Each link carries $U_e \in \text{SU}(2)$, and the configuration space carries the product

Haar probability measure. The physical Hilbert space is the subspace of L^2 invariant under the gauge transformations at every vertex, boundary vertices included.

Hamiltonian (the workbench's normalization):

$$H_L = \kappa K + \kappa \xi (2M_L - S), \quad K = -\sum_{\{e,a\}} X^2_{\{e,a\}}, \quad S = \sum_p W_p, \quad \kappa = 2g^2/a, \quad \xi = 1/(4g^4).$$

Here $X_{\{e,a\}}$ differentiates U_e along $e^{\{tT_a\}}$ with $T_a = -i\sigma_a/2$, so K is the sum of the link Casimirs, and W_p is the trace of the ordered product of the four links around p . The plaquette term equals $(2g^2a)^{-1} \sum_p (2 - W_p) \geq 0$. The gap Δ_L is the distance from the ground energy E_0 to the rest of the spectrum on the physical space. At $\xi = 0$ it equals 3κ : one plaquette in the fundamental representation, four links of Casimir $3/4$. Large g is strong coupling, where ξ is small. The workbench records the conversion $g_C = 2^{3/4}g$ to the convention of [SZH]; the thresholds below are in the workbench's normalization.

The approach. The ground state is positive, so it can be written $\psi = e^v / \|e^v\|$. The eigenvalue equation becomes $v = \xi v_1 + B(v, v)$ with $v_1 = S/3$ (Lemma 2.4), whose formal solution is $\sum \xi^n v_n$. The workbench computes $v_1, \dots, v_5$ exactly, bounds them in weighted Fourier-algebra norms that are local, so that nothing grows with the box, and solves for the remainder by a contraction; returning to the eigenvalue equation bounds every excitation energy from below (Proposition 2.5).

2. The strong-coupling gap

Theorem 2.1 (the fifth-reference gap bound; F26-F42). Let (m_i, t_i) , $i = 1, \dots, 5$, be the ten rationals of the workbench's F26, $\ell_5(x) = \sum (m_i + 4t_i)x^i$, $\delta_5(x) = \sum_{\{i,j \leq 5, i+j \geq 6\}} 3(m_{i-t_j} + m_{j-t_i})x^{i+j}$, $\mathcal{D}_5 = (1 - \ell_5)^2 - (8/3)\delta_5$, α_5 the first positive root of $\mathcal{D}_5$, and

$$d_5(x) = (3/2)(1 + \sqrt{\mathcal{D}_5(x)}) + \sum_{\{i \leq 5\}} ((3/2)m_i - 6t_i)x^i.$$

If the (m_i, t_i) bound the local norms of the vacuum coefficients, as the workbench's computations assert, then for every $L \geq 2$, every $a > 0$ and every $0 < \xi \leq \alpha_5$,

$$\Delta_L \geq \kappa d_5(\xi).$$

Here $\xi \leq \alpha_5 \iff g^2 \geq 1/(2\sqrt{\alpha_5}) = 3.6835519839857273\dots$, with $\alpha_5 = 0.0184249535761176166818\dots$; d_5 decreases on $[0, \alpha_5]$ from 3 to $d_5(\alpha_5) = 1.5453\dots$, so $\Delta_L \geq 1.5453\kappa$ for all $g^2 \geq 3.6836$, and since $d_5(1/64) = 1.9068301567\dots$, $\Delta_L > 1.9068\kappa$ for all $g^2 \geq 4$, uniformly in the box and the spacing.

Status. Conditional on the computed coefficient bounds of orders 3, 4 and 5 for the threshold 3.6836 and the constants d_5 . Order 1 is verified by hand. Order 2 was recomputed from explicit $SU(2)$ coefficient tensors: $m(v_2) = 134.0673\dots$ and $t(v_2) = 19.7953\dots$, below the workbench's 5834/39 and 137/6. Order 5 (662 representatives, 124,864 anchored multisets, 5,726 rational primal/dual certificates) was reproduced by re-running the workbench's producer and auditor, with byte-identical output; its logic was read, not re-implemented. Orders 3–4 are inherited from earlier continuations and pinned by hash. The analytic argument is verified; α_5 , the threshold and d_5 were recomputed by exact bisection and root isolation. The qualitative content holds with verified inputs for $g^2 \geq 3.9781$ (Corollary 2.6(b)).

The proof has two lemmas and an analytic core.

Lemma 2.2 (the gauge-invariance Casimir inequality; generalized here). Let the links form a finite graph without loops, parallel links allowed, in which every cycle has length at least g . On every Peter-Weyl block j that contains a nonzero gauge-invariant vector, $\sum_j j_f \geq g \cdot j_e$ for every link e . Hence $c_j \geq (3g/2)j_e$ for every e , $\sum_e j_e \leq (2/3)c_j$, and $c_j \geq 3g/4$ on the nonconstant physical space. For the cubic box ($g = 4$): $c_j \geq 6j_e$ and $c_j \geq 3$. The first inequality holds for all real weights $j_f \geq 0$ satisfying the vertex inequalities $j_f \leq \sum_{\{f \ni v, f \neq f\}} j_{\{f\}}$, and it is an equality for $j = 1/2$ on a cycle of length g through e and 0 elsewhere.

Proof. A block contains a nonzero invariant exactly when at every vertex the tensor product of the spins j_f ($f \ni v$) contains the trivial representation; by the Clebsch-Gordan rule this gives the vertex inequalities, which are all that is used.

A matching at each vertex. At v the inequalities say that no j_f exceeds half of $J_v = \sum_{\{f \ni v\}} j_f$. By Gale's supply-demand theorem this is the condition for a transport plan between the links at v with supplies and

demands j_f and the diagonal forbidden. Symmetrizing gives a symmetric $M^v \geq 0$ with zero diagonal and row sums j_f .

A circulation. Give each orientation (dart) of a link f the weight $j_f/2$, and let the dart arriving at v along f pass to the dart leaving v along $f' \neq f$ with weight $M^v_{ff'}/2$. This is a circulation on the darts, hence a nonnegative combination $\sum_C w_C |C|$ of closed walks in which consecutive links differ, with $\sum_C w_C |C| = \sum_f j_f$ and $\sum_C w_C n_e(C) = j_e$, where $n_e(C)$ counts the traversals of e .

Each walk. Cut a walk C with $n_e(C) \geq 1$ at its traversals of $e = uw$. Each piece avoids e and is nonempty. A piece that returns to its starting endpoint is a closed walk with consecutive links distinct, so it contains a cycle and has length at least g ; a piece joining u and w contains a path, which with e forms a cycle, so it has length at least $g - 1$. Hence $|C| \geq g \cdot n_e(C)$, and summing gives $\sum_f j_f \geq g \cdot j_e$.

The Casimir. Every nonzero spin is at least $1/2$, so $j_f(j_f + 1) \geq (3/2)j_f$, which gives $c_j \geq (3/2)\sum_f j_f$ and the remaining inequalities. The equality case is $j = 1/2$ on a cycle of length g . ■

The workbench's F7 is the case $g = 4$ on the cubic lattice, proved there from the absence of triangles. The inequality was checked exhaustively over half-integer spins with vertex singlets on the cube graph, $K_{3,3}$, the 5-cycle, the Petersen graph, the 6-cycle and a triangle, and by linear programming over real weights on thirteen graphs of girth 3 to 8 (among them the Petersen, Heawood, McGee and Tutte-Coxeter graphs, the dodecahedron and the box $\{-1, 0, 1\}^3$), on three parallel links and on the bridge between two triangles. On every link that lies on a shortest cycle the optimum is the girth, and on the bridge it is 4, larger than the girth 3.

Remark 2.3 (the exact constant for each link). For a link $e = uw$ let g_e be the length of a shortest cycle through e (∞ if none) and ℓ_x the length of a shortest closed walk at x in the graph without e in which consecutive links differ, except possibly the last and the first. Then $\sum_f j_f \geq \gamma_e j_e$ with $\gamma_e = \min(g_e, 1 + (\ell_u + \ell_w)/2) \geq g$, and γ_e is attained by half-integer spins with vertex singlets.

Proof. Cut each walk of the circulation at its traversals of e as above. A piece joining u and w has length at least $g_e - 1$; a piece returning to its starting endpoint x has length at least ℓ_x , and since C is closed the returning pieces come in pairs, one at u and one at w . Hence $|C| \geq n_e(C)\gamma_e$. For attainment, spin $1/2$ on a shortest cycle through e gives g_e ; for the other value take $C = eP_w e^{-1}P_u$ with P_w, P_u shortest walks of the kind defining ℓ_w, ℓ_u , and $j_f = n_f(C)/2$. ■

Lemma 2.4 (ground-state transform; F33-F36). Let $\Gamma(f, h) = \sum_{\{e,a\}} (X_{\{e,a\}} f)(X_{\{e,a\}} h)$, P_H the Haar mean and $Q_H = I - P_H$. For real or complex v , e^v is an eigenfunction of H_L with eigenvalue E if and only if $Kv = \xi S + Q_H \Gamma(v, v)$ and $E = \kappa(2M_L \xi - P_H \Gamma(v, v))$; then $(H_L - E)(e^v h) = \kappa e^v (Kh - 2\Gamma(v, h))$ for every h .

Proof. From $X^2(e^v h) = e^v (h(Xv)^2 + 2(Xv)(Xh) + hX^2v + X^2h)$ one gets $K(e^v h) = e^v (h(Kv - \Gamma(v, v)) - 2\Gamma(v, h) + Kh)$. With $h = 1$, $H_L e^v = E e^v$ exactly when $Kv - \Gamma(v, v) - \xi S$ is constant, and taking Haar means identifies the constant. The identities are algebraic, so they hold for complex v and ξ ; only the positivity of e^v , used for the ground state, needs v real. ■

With $v_1 = S/3$ (each W_p has Casimir 3) and $B = K^{-1}Q_H \Gamma$, the vacuum equation is $v = \xi v_1 + B(v, v)$, with formal solution $\sum \xi^n v_n$, $v_n = \sum_{\{i+j=n\}} B(v_i, v_j)$; this is the coupled-cluster ansatz of [SZH].

The analytic core (F4-F32; verified). In the Fourier-algebra norm $\|f\|_X = \sum_j \|A_j(f)\|_1$, which is submultiplicative [Ey], and in local weighted norms $m(\cdot)$, $t(\cdot)$, $\|\cdot\|_{\text{loc}}$ anchored at a link, one has $\|B(f, h)\|_{\text{loc}} \leq 3(m(f)t(h) + m(h)t(f)) \leq (2/3)\|f\|_{\text{loc}}\|h\|_{\text{loc}}$, by Lemma 2.2. With $q_5 = \sum_{\{i \leq 5\}} \xi^i v_i$, the residual $\xi v_1 + B(q_5, q_5) - q_5$ has degrees 6–10 and local norm at most $\delta_5(\xi)$, and $J_5 = 2B(q_5, \cdot)$ has norm at most $\ell_5(\xi)$. For $0 < \xi \leq \alpha_5$ a Neumann inverse and a binary-tree majorant give $v = q_5 + w$ with $\|w\|_{\text{loc}} \leq w^* = (3/4)(1 - \ell_5 - \sqrt{\mathcal{D}_5})$, the small root of $(2/3)w^2 - (1 - \ell_5)w + \delta_5 = 0$. The bounds are uniform in the box: each v_n is a sum of universal cluster functions over plaquette multisets, so the anchored sums of the infinite lattice bound every box, and the orientation dependence of the norm is handled by the maximum over the eight axis reflections.

Proposition 2.5 (spectral return; F35-F37). Under the analytic core, every eigenvalue $\lambda > 0$ of $H_L - E_0$ on the physical space satisfies $\lambda \geq \kappa \delta_5(\xi)$.

Proof. Let φ be a physical eigenvector, $h = \varphi/\psi$ and $h' = Q_H h \neq 0$. By Lemma 2.4, $(K - \lambda/\kappa)h' = 2Q_H \Gamma(v, h')$, and $\|Kh'\|_X$ is finite (F36). If $\lambda/\kappa \geq 3$ there is nothing to prove, since $d_5 \leq 3$. Otherwise $c_j \geq 3$ on every nonconstant physical block (Lemma 2.2), so $\|(K - \lambda/\kappa)h'\|_X \geq (1 - \lambda/(3\kappa))\|Kh'\|_X$. The core gives $\|2\Gamma(v, h')\|_X \leq 4t(v)\|Kh'\|_X \leq \chi\|Kh'\|_X$ with $\chi = 4\sum_i \xi_i + (2/3)w^*$, and Q_H does not increase $\|\cdot\|_X$. Hence $\lambda \geq 3\kappa(1 - \chi)$, and substituting w^* gives $3(1 - \chi) = d_5(\xi)$. ■

Nothing in this argument uses the truncation order, which gives the following.

Corollary 2.6 (the gap at every truncation order; generalized here). For $1 \leq N \leq 5$ define $\ell_N, \delta_N, \mathcal{D}_N, \alpha_N$ and d_N as above with 5 replaced by N and the residual degrees $i + j \geq N + 1$. If (m_i, t_i) , $i \leq N$, bound the local norms of the first N vacuum coefficients, then $\Delta_L \geq \kappa d_N(\xi)$ for every $L \geq 2$, $a > 0$ and $0 < \xi \leq \alpha_N$, and d_N decreases on $[0, \alpha_N]$. In particular:

- (a) $N = 1$: $\mathcal{D}_1(\xi) = 1 - 256\xi/3$ exactly and $(3/2)m_1 - 6t_1 = 0$, so $\Delta_L \geq (3/2)\kappa(1 + \sqrt{1 - 64/(3g^4)})$ for all $g^2 \geq 8/\sqrt{3} \approx 4.6188$, with 2.0745κ at $g^2 = 5$ and 2.8304κ at $g^2 = 10$.
- (b) $N = 2$: with the workbench's pair $(m_2, t_2) = (5834/39, 137/6)$, the threshold is $g^2 \geq 4.0186791538833\dots$ with $\Delta_L \geq 1.52094\kappa$. With the recomputed values rounded up to $(134.0674, 19.7954)$, the threshold is $g^2 \geq 3.97807\dots$ with $\Delta_L \geq 1.52054\kappa$ on that range and $\Delta_L \geq 1.64707\kappa$ at $g^2 = 4$.
- (c) $N = 4$: $g^2 \geq 3.7159603625\dots$, with $d_4(1/64) = 1.898118$.

Proof. Replace q_5 by q_N in the core. The residual is $\sum_{\{i,j \leq N, i+j \geq N+1\}} \xi^{i+j} B(v_i, v_j)$, of local norm at most δ_N , and $\|2B(q_N, \cdot)\| \leq \ell_N$. On $[0, \alpha_N]$, $\ell_N < 1$, since $\ell_N(\xi_1) = 1$ would give $\mathcal{D}_N(\xi_1) < 0$. The Neumann inverse and Proposition 2.5 give $\lambda \geq \kappa d_N(\xi)$, and differentiating the quadratic for w^* gives $w^* \sqrt{\mathcal{D}_N} = \delta_N' + \ell_N' w^* > 0$, so d_N decreases. For $N = 1$, $(m_1, t_1) = (64/3, 16/3)$ gives the closed form. ■

The recomputed order-2 values have closed forms $m(v_2) = 6 + 11200/117 + 112/9 + 448\sqrt{3}/39$ and $t(v_2) = 3/2 + 80/9 + 256/39 + 64\sqrt{3}/39$, which agree with the tensor computation to ten digits. So the qualitative content of Theorem 2.1, a gap on the whole physical space uniform in the box and the spacing, holds with verified inputs for $g^2 \geq 3.9781$, which includes the benchmark $g^2 = 4$; orders 3–5 lower the threshold to 3.6836 and raise the constants. Part (a) is the workbench's own order-1 theorem of its 14–16 September continuation, and (b) strengthens its single-norm order-2 theorem (threshold ≈ 4.1156).

Corollary 2.7 (the gap on the full space; new). Under the hypotheses of Corollary 2.6, every eigenvalue $\lambda > 0$ of $H_L - E_0$ on the whole space L^2 , not only the physical space, satisfies $\lambda \geq (1/4)\kappa d_N(\xi)$. This is sharp at $\xi = 0$, where the gap on L^2 is $(3/4)\kappa$: one link in spin 1/2.

Proof. The positive ground state is unique on L^2 as well. Lemma 2.4 holds for every h , and the bound $\|2\Gamma(v, h)\|_X \leq 4t(v)\|Kh\|_X$ uses only $\sum_e j_e \leq (2/3)c_j$, which holds on every block. On L^2 the nonconstant blocks have $c_j \geq 3/4$ instead of 3, so the proof of Proposition 2.5 gives $\lambda \geq (3/4)\kappa(1 - \chi) = (1/4)\kappa d_N$. ■

Proposition 2.8 (the gap at complex coupling; new). Suppose the inputs of Corollary 2.6 are valid through order N , and let $\zeta \in \mathbb{C}$ with $|\zeta| \leq \alpha_N$. For every $L \geq 2$, on the physical space:

1. $H_L(\zeta)e^{\{v(\zeta)\}} = E_0(\zeta)e^{\{v(\zeta)\}}$ with $E_0(\zeta) = \kappa(2M_L\zeta - P_H\Gamma(v, v))$, analytic in $|\zeta| < \alpha_N$;
2. every other eigenvalue $E_0(\zeta) + \lambda$ of $H_L(\zeta)$ has $\text{Re } \lambda \geq \kappa d_N(|\zeta|)$;
3. $E_0(\zeta)$ is algebraically simple.

Hence the ground-state Riesz projection of $H_L(\zeta)$ is analytic in $|\zeta| < \alpha_N$, for every L .

Proof. (1) The fifth-reference manuscript states the core for complex ξ with $|\xi| \leq \alpha$, and Lemma 2.4 holds for complex v ; the v_n and w are analytic in ζ , being locally uniform limits of polynomials. (2) For an eigenvector with eigenvalue $E_0 + \lambda$, $\lambda \neq 0$, put $h' = Q_H(e^{\{-v\}}\varphi) \neq 0$; then $(K - \lambda/\kappa)h' = 2Q_H\Gamma(v, h')$. On blocks with $c_j \geq 3$, $|c_j - \lambda/\kappa| \geq c_j(1 - \text{Re } \lambda/(3\kappa))$ if $\text{Re } \lambda \geq 0$, and $|c_j - \lambda/\kappa| \geq c_j$ if $\text{Re } \lambda < 0$. With $\chi < 1$ the case $\text{Re } \lambda < 0$ is impossible, and otherwise $\text{Re } \lambda \geq 3\kappa(1 - \chi)$. (3) With $\lambda = 0$ the same argument gives $h' = 0$; a generalized eigenvector $e^{\{v\}}h$ with $(H_L - E_0)(e^{\{v\}}h) = e^{\{v\}}v$ would give $Kh' = 2Q_H\Gamma(v, h')$, so $h' = 0$ and then $0 = 1$. The family $H_L(\zeta)$ is holomorphic of type (A) with compact resolvent, and the simple eigenvalue is separated by a strip, so its Riesz projection is analytic [Ka, Ch. VII]. ■

By contrast, the global perturbative route certifies only $|\zeta| < 3/(4M_L)$, at most $1/320$ (Proposition 6.1).

The proof of Theorem 2.1 uses the Hamiltonian, its Fourier calculus and the computed coefficients. The Jacobian, Navier-Stokes and S^6 inputs of the workbench enter other results, described in §5.

3. Heat flow and the complementary physical space

The second manuscript of the latest edition (H1-H34) uses the control of the vacuum for two further purposes: box-independent bounds on the heat correlations of plaquettes, and bounds on how much the plaquette family mixes with the rest of the physical space.

Objects. Let $A = H_L - E_0$ on the centred physical space and $\rho = \psi^2$. The centred plaquette states and their heat correlations are $r_p = (W_p - \langle W_p \rangle_\rho) \psi$ and $\hat{C}_{pq}(\tau; \xi) = \langle r_p, e^{\{-\tau A/\kappa\}} r_q \rangle$, $\tau = \kappa t$. The row norm is $\|B\|_{\text{row}} = \max_p \sum_q |B_{pq}|$. C_0, C_2, C_4 are the coefficients of ξ^0, ξ^2, ξ^4 in $\hat{C}$, computed by the workbench. In the coordinates $f \mapsto \psi f$, Lemma 2.4 turns A/κ into $K - D_v$ with $D_{vh} = 2\Gamma(v, h)$, and $\|D_{vh}\|_X \leq \chi \|Kh\|_X$ with χ as in Proposition 2.5, so $3(1 - \chi) = d_5(\xi)$.

Theorem 3.1 (heat remainder; H13, prefactor improved here). For every $L \geq 2$, every $\tau \geq 0$ and every $|\xi| < 1/55$,

$$\|\hat{C}(\tau; \xi) - C_0(\tau) - \xi^2 C_2(\tau) - \xi^4 C_4(\tau)\|_{\text{row}} \leq (2829/13) e^{\{-13\tau/8\}} (55|\xi|)^6 / (1 - (55|\xi|)^2).$$

The workbench states this with the prefactor $67896/169 = 401.75\dots$ in place of $2829/13 = 217.61\dots$; its bound is also valid.

Status. Conditional on the analytic core of Theorem 2.1, hence on the same order-3 to order-5 inputs. Every constant is re-derived below; the semigroup construction (H5-H7) and the continuation to complex ξ were read, and Proposition 2.8 gives the spectral side of that continuation.

Proof. (i) *The mean and the semigroup.* Put $T = Q_{HD} v K^{-1}$ and $p = P_{HD} v K^{-1}$ on the nonconstant physical coefficients. Since $\|\cdot\|_X$ adds the scalar and the nonconstant parts, $|py| + \|Ty\|_X \leq \chi \|y\|_X$, so $S = (I - T)^{-1}$ exists with $\|S\| \leq 1/(1 - \chi)$, and the mean is $\mu(F) = P_{HF} + pSQ_{HF}$. The additive form of the inequality gives $|(\mu - P_H)F| \leq \chi \|Q_{HF}\|_X$, sharper than the workbench's $\chi \|Q_{HF}\|_X / (1 - \chi)$: with $y = Q_{HF}$ and $z = Sy$ one has $z = y + Tz$, so $|pz| \leq \chi \|z\|_X - \|Tz\|_X \leq \chi \|y\|_X - (1 - \chi) \|Tz\|_X \leq \chi \|y\|_X$. For $L_0 = (I - T)K$ and $h > 0$, $\|(I + hL_0)f\|_X \geq (1 + 3h(1 - \chi))\|f\|_X$ by Lemma 2.2, and with the range condition (H5) the Lumer-Phillips theorem [LP] gives $E(\tau) = e^{\{-\tau L_0\}}$ with $\|E(\tau)\| \leq e^{\{-3(1 - \chi)\tau\}}$.

(ii) *A formula for the correlations.* With $h_\tau = K^{-1}SE(\tau)W_p$ and $G_z = \sum_q z_q W_q$, $|z_q| \leq 1$, the Poisson equation (H3) and one integration by parts give $\sum_q z_q \hat{C}_{pq}(\tau) = \mu\Gamma(h_\tau, G_z)$, with $\|Kh_\tau\|_X \leq (8/(1 - \chi))e^{\{-3(1 - \chi)\tau\}}$.

(iii) *Evenness.* The map $U_{\{(n,i)\}} \mapsto (-1)^{\{\sum_{k < i} n_k\}} U_{\{(n,i)\}}$ multiplies each link by a central element, preserves the Haar measure, K and the gauge action, and sends every W_p to $-W_p$. It conjugates $H_L(\xi)$ to $H_L(-\xi)$ plus a constant, so $\hat{C}(\tau; -\xi) = \hat{C}(\tau; \xi)$ and only even powers occur (checked on all 240 faces of the box $L = 2$).

(iv) *Cauchy's estimate.* By Proposition 3.2 below, on the circle $|\zeta| = 1/55$ every phase-weighted row sum of $\hat{C}(\tau; \zeta)$ is at most $(2829/13)e^{\{-13\tau/8\}}$; so the coefficient of ζ^{2n} has row norm at most $(2829/13)e^{\{-13\tau/8\}} 55^{2n}$, and summing over $n \geq 3$ gives the tail. ■

Proposition 3.2 (the constants; H9-H12).

1. $\|\Gamma(h, G_z)\|_X \leq 32\|Kh\|_X$.
2. $|P_H\Gamma(h, G_z)| \leq (1/8)\|Kh\|_X$.
3. $\|\hat{C}(\tau)\|_{\text{row}} \leq ((1 + 255\chi)/(1 - \chi)) e^{\{-3(1 - \chi)\tau\}}$, and a fortiori the workbench's $(1 + 255\chi)/(1 - \chi)^2 e^{\{-3(1 - \chi)\tau\}}$.
4. At $|\xi| = 1/55$ one has $1/55 < \alpha_5$ and $d_5(1/55) = 1.637\dots > 13/8$, so $\chi < 11/24$; the prefactor is at most $(1 + 255 \cdot 11/24)/(13/24) = 2829/13$ and the rate at least $13/8$.

Proof. (1) By (F10), $\|\Gamma(h, G)\|_X \leq 2t(G)\|Kh\|_X$. W_q has one Fourier block, spin $1/2$ on the four links of q , with coefficient trace norm $2^3 = 8$ (F12); a link lies on at most four plaquettes, so $t(G_z) \leq 4 \cdot (1/2) \cdot 8 = 16$. (2) Integration by parts gives $P_H\Gamma(h, G_z) = 3\sum_q z_q q f h W_q$, and the component a_q of h on the

line CW_q contributes $3 \cdot 8|a_q|$ to $\|Kh\|_X$. (3) $\mu\Gamma = P_H\Gamma + (\mu - P_H)Q_H\Gamma$, and step (i), (2) and (1) give $|\mu\Gamma(h_\tau, G_z)| \leq ((1 - \chi)/8 + 32\chi)\|Kh_\tau\|_X \leq ((1 + 255\chi)/(1 - \chi))e^{-3(1-\chi)\tau}$. (4) is checked in exact rational arithmetic. ■

Other radii (generalized here). Nothing in the proof uses $R = 1/55$ except the numbers of Proposition 3.2(4). For every $0 < R \leq \alpha_5$ the same proof gives Theorem 3.1 with $55|\xi|$ replaced by $|\xi|/R$, the prefactor by $C(R) = (1 + 255\chi_R)/(1 - \chi_R)$ and the rate by $d_5(R)$, where $\chi_R = 1 - d_5(R)/3$. At $R = 1/55$ the exact $\chi_R = 0.45412\dots$ gives the prefactor $213.97\dots$ and the rate $1.6376\dots$; at $R = \alpha_5$ the prefactor is $241.99\dots$ and the rate $1.5453\dots$; at $R = 45/4096$, the radius of the earlier edition's bound U29b, the prefactor is $85.00\dots$ and the rate $2.2588\dots$, which supersedes U29b (prefactor $6184/25$, rate $15/8$). At $\xi = 0$ one has $C_0(\tau) = e^{-3\tau}I$ exactly.

Relaxation into the whole complementary space. Let R be the map with columns r_p and $\Phi = \kappa A^{-1}R$, with Grams $G_0 = R^*R$, $G_1 = R^*\Phi$, $G_2 = \Phi^*\Phi$, energy $E = \kappa G_1$ and $K_o = \kappa^{-1}R^*AR$. Put $P = \Phi G_2^{-1}\Phi^*$, $Q = I - P$, $B_\Phi = QR$ and $D_Q = QAQ$ on $Q\mathcal{H}_o$. Allowing the family Φc to relax into every $h \in Q\mathcal{H}_o$ gives $E_{\text{full}} = E - \kappa^2 B_\Phi^* D_Q^{-1} B_\Phi$ and $G_{\text{full}} = G_2 + \kappa^2 B_\Phi^* D_Q^{-2} B_\Phi$.

Proposition 3.3 (relaxation bounds; H20-H29). Suppose $\|G_k - 3^{-k}I\| \leq w_k$ ($k = 0, 1, 2$), $\|K_o - 3I\| \leq w_K$, $0 \leq J = G_0 - 6G_1 + 9G_2 \leq \beta I$ and $A \geq \kappa d_{\text{ph}}$. Put $l_k = 3^{-k} - w_k > 0$, $u_k = 3^{-k} + w_k$ and $u_K = 3 + w_K$. Then:

1. the fraction of each Φc seen by the plaquette states is at least $\max\{l_1^2/(u_0 u_2), 1 - \beta/(9l_2)\}$;
2. $(1 - \eta_E)E \leq E_{\text{full}} \leq E$ and $G_2 \leq G_{\text{full}} \leq (1 + \eta_G)G_2$, with $\eta_E = \beta/(d_{\text{ph}} l_1)$ and $\eta_G = \beta/(d_{\text{ph}}^2 l_2)$;
3. the energy of $(I - P_R)\Phi c$ is at most $(u_1 u_K / l_0^2 - 1)$ times $c^* E c$.

Proof. (1) $\|P_R \Phi c\|^2 = c^* G_1 G_0^{-1} G_1 c \geq (l_1^2/u_0)|c|^2$ and $\|\Phi c\|^2 \leq u_2|c|^2$. For the second bound (new here): G_1 is symmetric, so $J = (R - 3\Phi)^*(R - 3\Phi)$ and $\|(I - P_R)\Phi c\|^2 \leq \|\Phi c - Rc/3\|^2 \leq \beta|c|^2/9$, while $\|\Phi c\|^2 \geq l_2|c|^2$. (2) $B_\Phi^* B_\Phi = G_0 - G_1 G_2^{-1} G_1$ is the least value of $(R - \Phi X)^*(R - \Phi X)$, and $X = 3I$ gives J ; with $D_Q \geq \kappa d_{\text{ph}}$ the bounds follow. (3) Since $A\Phi = \kappa R$, the mixed energy is $\kappa c^* G_1 c$, and expanding gives $\kappa c^*(G_1 G_0^{-1} K_o G_0^{-1} G_1 - G_1)c \leq (u_K u_1 / l_0^2 - 1)\kappa c^* G_1 c$. ■

Proposition 3.4 (the kinetic rows; H14, derived here). For each plaquette p :

1. $\Gamma(W_p, W_p) = 3 - \chi_1(U_p)$, of norm $3 + 27 = 30$;
2. for each of the 12 plaquettes q adjacent to p , $\Gamma(W_p, W_q) = (3/4)P_0 - (1/4)P_1$, where P_0, P_1 are the parts of $W_p W_q$ with spin 0 and 1 on the shared link; $\|\Gamma(W_p, W_q)\|_X$ is 24 for 8 of the q and $6 + 6\sqrt{3}$ for the other 4, both below the workbench's 48; for non-adjacent $q \neq p$, $\Gamma(W_p, W_q) = 0$;
3. hence $\sum_q |\mu\Gamma(W_p, W_q)| \leq 3 + \chi(27 + 192 + 24 + 24\sqrt{3}) < 134$ for $\chi < 11/24$, against the workbench's 606.

Proof. From $2\Gamma(f, h) = (c_f + c_h)fh - K(fh)$ for block eigenfunctions (F17): $W_p^2 = 1 + \chi_1(U_p)$, and $\chi_1(U_p)$ has Casimir 8 and norm $3^3 = 27$; for adjacent faces $2\Gamma = 6(P_0 + P_1) - (9/2)P_0 - (13/2)P_1$, and the trace norms of P_0, P_1 are (16, 48) for two thirds of the adjacent pairs and (8, $24\sqrt{3}$) for the rest (computed from explicit $SU(2)$ tensors). Non-adjacent faces share no link. ■

Proposition 3.5 (the benchmark tables; H5). From the workbench's inherited degree-2 and degree-4 row bounds and the constants $C = 67896/169$, C_J and 606, Proposition 3.3 gives, for every $L \geq 2$ and $a > 0$:

Coupling	Observed fraction	Energy loss η_E	Gram increase η_G
$g^2 \geq 10$	> 0.9778	< 0.01057	< 0.01123
$g^2 \geq 12$	> 0.9975	< 0.00115	< 0.00120
$g^2 \geq 25/2$	> 0.9984	< 0.00071	< 0.00073
$g^2 \geq 13$	> 0.99904	< 0.000436	< 0.000451
$g^2 \geq 16$	> 0.99991	< 0.0000357	< 0.0000364

This reproduces the workbench's table: at $g^2 \geq 13$ the energy loss and the Gram increase are below $1/2000$, and at $g^2 \geq 16$ below $1/25000$. With the sharper constants of this record (the prefactor $2829/13$,

the leakage prefactor $C_J \cdot 13/24 = 782.29\dots$, the kinetic bound 134 and the second bound of Proposition 3.3(1)) the same computation gives:

Coupling	Observed fraction	Energy loss η_E	Gram increase η_G
$g^2 \geq 10$	> 0.99458	< 0.005713	< 0.006053
$g^2 \geq 12$	> 0.99940	< 0.000623	< 0.000647
$g^2 \geq 25/2$	> 0.99963	< 0.000380	< 0.000394
$g^2 \geq 13$	> 0.99977	< 0.000237	< 0.000244
$g^2 \geq 16$	> 0.999981	< 0.0000194	< 0.0000198

Status. Recomputed in exact rational arithmetic from the stated inputs; at $g^2 = 13$, with the workbench's $d_{ph} = 29047/10000$, the first table agrees with its diagnostics to ten digits. Conditional on the inherited row bounds and on the inputs of Theorem 2.1. The widths are monotone in ξ , so each row holds for all larger g^2 .

Proposition 3.6 (an upper bound on the gap; new).

1. For every $L \geq 2$, $a, g > 0$ and every plaquette p , $\Delta_L \leq \kappa(3 - \langle \chi_1(U_p) \rangle_p) / (1 + \langle \chi_1(U_p) \rangle_p - \langle W_p \rangle_p^2)$.
2. With the widths of Proposition 3.3, $\Delta_L / \kappa \leq (3 + w_K) / (1 - w_o)$; with the constants above the gap lies in $[2.8382, 3.0055]\kappa$ for $g^2 \geq 10$, in $[2.9047, 3.00024]\kappa$ for $g^2 \geq 13$ and in $[2.9372, 3.00003]\kappa$ for $g^2 \geq 16$, the lower ends being $d_5(\xi)$ under the inputs of Theorem 2.1.
3. At each fixed L the first-order term of Δ_L at $\xi = 0$ vanishes, and $\Delta_L \leq 3\kappa + O(\xi^2)$ uniformly in L ; every lower bound d_N is $3 - 64\xi + O(\xi^2)$, so its first-order loss is a feature of the method.

Proof. (1) r_p is gauge invariant and orthogonal to ψ ; by min-max, $\Delta_L \leq q_A(r_p) / \|r_p\|^2$, with $q_A(r_p) = \kappa(3 - \langle \chi_1 \rangle_p)$ by Proposition 3.4(1) and $\|r_p\|^2 = 1 + \langle \chi_1 \rangle_p - \langle W_p \rangle_p^2$. (2) The two quantities are $(K_o)_{pp} \leq 3 + w_K$ and $(G_o)_{pp} \geq 1 - w_o$. (3) At $\xi = 0$ the eigenvalue 3κ has eigenspace spanned by the W_p (by Lemma 2.2, $c_j = 3$ only for spin 1/2 on a 4-cycle); the first-order matrix $\langle W_q, SW_{\{q'\}} \rangle$ vanishes because in each product $W_q W_{r'} W_{\{q'\}}$ of faces some link occurs exactly once in spin 1/2, and $\int W_q^3 = 0$; the ground energy has no first-order term since $\int S = 0$; and $d_N'(0) = -64$. ■

Heat horizons, support quotients and the spatial limit. Replacing Φ by $\Phi_T = (I - e^{\{-TA\}})\Phi$ changes the Grams by factors between $(1 - \varepsilon)^2$ and 1 with $\varepsilon = e^{\{-\kappa d_{ph} T\}}$; for a signed sum of log det terms of total rank B the error is at most $2B\varepsilon/(1 - \varepsilon)$, and $T = (\kappa d_{ph})^{-1} \log(1 + 2B/\eta)$ makes it at most η (H33; the choice of T checked here). The support quotients (H31–H32) and the spatial limit at fixed spacing (H34, an outline) are not verified here; H34 builds, from the box-independence of Theorem 3.1, the infinite-volume heat semigroup, its invariant state and the relaxation bounds at fixed a and g .

4. The earlier editions

The gap and heat bounds of the continuations of 14–21 September are superseded by §§2–3; their other calculations remain in the record as dated editions.

- **14–16 September** (the 297-page web continuation). A gauge-native order-1 theorem (G22, G26–G29) with $\Delta_L > 2.07445\kappa$ at $g^2 \geq 5$, which is Corollary 2.6(a); a single-norm order-2 theorem (R10) for $g^2 \geq 4.1156$, strengthened by Corollary 2.6(b); and an order-2 theorem (L17) for $g^2 \geq 3.8260$, which uses a computed bound $944984/351$ for the cubic term of the residual that is not verified here. One arithmetic line of this edition justifies its self bound $40/27$ by “ $38/81 + 1564/810 = 40/27$ ”, whose left side equals $12/5$; the value $40/27 = 3 \cdot (1/81) \cdot 8 + 15 \cdot (1/810) \cdot 64$ is correct for the self cluster, so the bound stands.
- **17 September** (the 49-page quartic cube reader). The fourth-order vacuum source in two presentations, equal as polynomials (78 representatives, 8,621 anchored multisets); the sixth-order vacuum energy with the cube contribution $-83/1944$; and $\Delta_L > 1.6584\kappa$ for $g^2 \geq 15/4$, which is Corollary 2.6(c) at $\xi = 4/225$ (Theorem 2.1 gives $d_5(4/225) > 1.6993$ there).
- **18–19 September.** The sixth vacuum source, the eighth-order energy and a response estimate on the 240 faces of the box $L = 2$; the workbench records that the new spatial tables and the final execution package were not supplied. Not verified here.

- **21 September, heat and volume** (the 46-page reader). The fourth-order heat coefficients (84 time functions over 17 geometric cases), a heat remainder on $|\xi| < 3/256$, the spatial limit at fixed spacing, and the full-complement energy retained above 999/1000 at $g^2 \geq 16$; superseded quantitatively by §3.

The foundations of 8-9 September. Three manuscript lines contain the workbench's first results, stated for the full finite Hamiltonian with open boundaries and all gauge constraints. The quantum line (79 pages) has exact extensive blocking maps, volume-independent lower bounds on the high-energy spectral weight of selected dressed Wilson-loop channels, a first volume-uniform gap $\Delta \geq \kappa(3 - 512\xi)$ for $0 < \xi \leq 1/49152$ (its §10, proved there with the creation method of [Ya]), a nonzero non-Abelian vertex on the box $L = 2$, and the Jacobian-map and Navier-Stokes branches. The spatial line (129 pages) studies sequences in which box, spacing and coupling vary together. The volume line (81 pages) has local electric moments at every coupling, the exact free physical threshold 3κ , and estimates on the S^6 -derived cusp at $0 < \xi \leq 10^{-16}$. Apart from the items treated in this record, these lines are not verified here.

Proposition 4.1 (the first volume-uniform gap is implied). For $0 < \xi \leq 3/256$, $d_1(\xi) \geq 3 - 128\xi$. So the quantum line's $\Delta \geq \kappa(3 - 512\xi)$ on $0 < \xi \leq 1/49152$ follows from Corollary 2.6(a), which holds on the larger range $\xi \leq 3/256$ ($g^2 \geq 8/\sqrt{3}$).

Proof. With $y = 256\xi/3 \in (0, 1]$, $d_1(\xi) = (3/2)(1 + \sqrt{1 - y}) \geq (3/2)(2 - y) = 3 - 128\xi$, since $\sqrt{s} \geq s$ on $[0, 1]$. ■

5. The typed maps from the side branches

Three branches of the workbench bring material from outside lattice gauge theory into the Hamiltonian setting, each through an explicit map. This section states each map and what is established about it.

The Jacobian map → local-energy trial states. The map F of Alpöge's counterexample ($\det DF = -2$, three preimages of $(-1/4, 0, 0)$; see the companion record on the counterexample) generates an incompressible polynomial flow, from which the workbench builds a material tensor, lattice local-energy weights and trial states, ending in its Theorem 14.2. That theorem is a valid diagonal argument, conditional on fixed-box weak-coupling limits that the file sketches. Its energy quotients equal, up to a factor $1 \pm 1/(10j)$, the lowest free two-gluon colour-singlet energy of an open box of side $j/50$ at couplings $g_j < 1/j$. So the map carries the Jacobian flow to trial states whose energies track the free two-gluon threshold, and the data of F enter the quotients within that window.

Navier-Stokes → an abelian connection. The velocity field of a solution becomes a reducible $SU(2)$ connection $A_i = \lambda u_i T$, with curvature density $-2\Sigma_{\{i<j\}} \text{tr } F_{ij}^2 = \lambda^2 |\text{curl } u|^2$, and gauge-invariant curvature states built from it are followed along a weak-coupling diagonal (quantum line §§20–22). The identity and the constants of the curvature masses are verified. The zero-quotient diagonal exists along couplings $g \rightarrow 0$ and rests on rate bounds for the Navier-Stokes solution; those rates are derived, conditionally on four named statements of OpenAI's manuscript, in the companion record on the Navier-Stokes workbench (its Theorem 3.1). The manuscript's main theorem is not verified here.

S^6 → a magnetic background. The spatial and volume lines use the imaginary-period block of the regular-fibre period matrix of the S^6 construction $[A]$ to build a magnetic background, magnetic link transports and a gauge-covariant map into the Wilson Hilbert space. On the patch these lines use, the period data enter through the single positive number $D = L_T q_T + 6m_T^2$, with $L_T = \text{Im } \tau_T$, $q_T = -\text{Im } \beta_T$ and $m_T = \text{Im } \mu_T$, through the angle $\theta = 2\pi a^2/D$. The global statement of $[A]$ is not verified here; the monodromy of the same period system is the subject of the companion record on the S^6 monodromy.

Zeta → the heat edition. The zeta programme's Gram-sandwich lemma, $(1 - \delta)^2 \Phi * \Phi \leq \Phi_J * \Phi_J \leq (1 + \delta)^2 \Phi * \Phi$ when $\|(\Phi_J - \Phi)x\| \leq \delta \|\Phi x\|$ and $\delta \leq 1$, is used in the 21 September heat edition with $\delta = e^{-T\lambda_0}$ from the heat semigroup; its steps are verified, and the condition $\delta \leq 1$ is necessary for the lower bound.

Collatz → a regression fixture. The identity $X_v - X_u = (q - 3)(q + 4)/32$ is used as a regression fixture; it is verified.

Part B. Proved limits of specific routes

6. The global perturbative route and the continuum path

Proposition 6.1 (the collapse of the global analyticity disk; spatial line §27, generalized here).

Write $H = (2g^2/a)(H_0 - \xi W_L) + \text{const}$ on the full space, with $H_0 = \sum_e E_e$ the sum of the link Casimirs and $W_L = \sum_p W_p$. The operator-norm Neumann series certifies the isolated rank-one ground-state projection on the circle $|z| = 3/8$ exactly for $|\xi| < 3/(16M_L) = 1/(64L^2(2L + 1))$, and on no contour separating 0 from the rest of the spectrum does the norm-product criterion give more. On the physical space the same route gives exactly $|\xi| < 3/(4M_L)$, on the circle $|z| = 3/2$.

Proof. $|W_p| \leq 2$ with equality at $U \equiv I$, so $\|\xi W_L\| = 2|\xi|M_L$, also on the physical space (the indicator of the gauge-invariant set $\{W_L > 2M_L - \varepsilon\}$ is a physical vector). The spectrum of H_0 lies in $\{0\} \cup [3/4, \infty)$, so $\|(H_0 - z)^{-1}\| \leq 8/3$ on $|z| = 3/8$, and both factors are attained. Any separating contour crosses $(0, 3/4)$, where the resolvent norm is at least $8/3$. On the physical space the spectrum of H_0 lies in $\{0\} \cup [3, \infty)$ by Lemma 2.2, which gives the resolvent norm $2/3$ on $|z| = 3/2$ and the radius $3/(4M_L)$. ■

The radius shrinks with the box, so at any fixed coupling this route eventually fails. It does not bound the true analyticity radius, which by Proposition 2.8 is at least α_N uniformly in the box under the inputs of Corollary 2.6. The comparison explains why the workbench's proof uses local norms: at $L = 2$ the global route certifies only $|\xi| < 1/1280$ on the full space and $|\xi| < 1/320$ on the physical space, far inside $\alpha_5 \approx 1/54.3$.

Proposition 6.2 (the standing continuum path leaves the domain). Write $c = 1/g^2$, so $\xi = c^2/4$ and $\xi \leq \alpha_5$ exactly when $c \leq 2\sqrt{\alpha_5} = 0.27147\dots$. On the workbench's standing path $a_n = a_0 2^{-n}$, $c_n = g_0^{-2} + \beta n \log 2$ with $\beta > 0$, the condition fails for every $n > (2\sqrt{\alpha_5} - g_0^{-2})/(\beta \log 2)$; so at most $1 + \lfloor (2\sqrt{\alpha_5} - g_0^{-2})/(\beta \log 2) \rfloor$ points of the path lie in the domain, and every point in the domain has spacing $a_n \geq a_0 e^{-2\sqrt{\alpha_5}/\beta}$.

Proof. $c \mapsto c^2/4$ is increasing, and c_n increases linearly without bound; a point in the domain has $n \leq 2\sqrt{\alpha_5}/(\beta \log 2)$. ■

So the estimates of §§2–3 hold along the path only down to a spacing bounded below in terms of a_0 and β ; the workbench states the first part itself (§H7).

Further limits recorded by the workbench. Each of the following is stated in the workbench, and each concerns one method, sequence or observable:

- re-centring the elementary order-1, single-norm certificate to extend the source domain returns its old endpoint $3/256 = \alpha_1$ (§H7);
- lower bounds for dressed channels are not bounds for the whole spectral measure (quantum line §11);
- along the logarithmic coupling path the upper end of the certified spectral window of the weighted covariance diverges, which the document calls a limitation of its own estimate (volume line §26);
- the zero-quotient sequence of the Navier–Stokes branch exists only along couplings tending to 0 (§5).

In the passages read, each is stated by the workbench with the scope of its proof.

Part C. Directions

7. The open directions, and my assessment

The workbench explored its problem the way the other programmes did: many routes, tested quickly with AI systems, some of which produced theorems and some of which mapped out what a route needs. This part states, for each open direction, what is established and my assessment.

Locally gapped, globally gapless sequences. The spatial line (129 pages) builds sequences in which the box, the spacing and the coupling vary together, driven by the S^6 period data through the magnetic

background of §5. Its conclusion (§28) states three results: along a fixed-coupling diagonal with $g_0^4 \geq 12288$ the magnetic excitation probability eventually has zero weight on every bounded energy interval; for fixed Wilson-loop states there is a positive high-energy mass and a failure of strong continuity; and on a selected weak-coupling diagonal the finite gaps close while the corrected magnetic probability escapes to high energy (§§17–18), with a unique-vacuum realization of a selected macroscopic limit of a second observable (§20). These are statements of the workbench and are not verified here. The aim of the direction, as the author describes it, is an object that has a gap at every finite regulator but whose continuum limit has none, because in the limit the spectrum becomes continuous; the S^6 period data supply the background that drives the sequences, and the author proposes extending the construction to higher-dimensional period data related to the Leech lattice (not verified here). My assessment: the mechanism the spatial line exhibits, finite gaps that close along a joint limit while spectral weight escapes, is the phenomenon that separates a lattice gap from a continuum gap, so the direction is well posed. Its next requirement is a sequence of this kind whose limit satisfies the reconstruction axioms under which the continuum problem is posed [JW, §4], since the present sequences are selected diagonals (Question 7).

Toward weak coupling. Every result of the workbench holds at a fixed lattice spacing and strong bare coupling, and Proposition 6.2 says how far along the standing path the estimates reach. The spatial line’s conclusion names the interacting local continuum and a nonzero low-energy spectral weight as its target (§28). My assessment: I think the most transferable ingredients of the present proof are Lemma 2.2, which is purely combinatorial and holds for every coupling, and the ground-state transform of Lemma 2.4, which is exact; what is specific to strong coupling is the convergence of the vacuum expansion in local norms. A route toward weak coupling would have to replace that expansion by a multiscale control of the vacuum, and the workbench’s blocking maps of the quantum line are a natural starting point for that.

Lower thresholds. The workbench names the same-support linearized action J_5 with a preconditioned residual as its next candidate. For certificates built only from majorant series with nonnegative coefficients, Pringsheim’s theorem suggests that re-centring cannot pass α_N ; the workbench’s calculation shows this for $N = 1$. Whether signed or same-support information can pass it is Question 2.

Independent confirmation of the inputs. An implementation independent of the workbench’s programs would remove the condition in Theorem 2.1. An independent referee computed the order-3 norms numerically with its own implementation (Clebsch–Gordan contraction over 612 anchored labels in 22 symmetry classes): $m(v_3) = 1203.1424\dots$ and $t(v_3) = 127.2886\dots$, below the workbench’s 1611.59 and 180.35, with the record’s energy coefficient $e_4 = 5M/216 - 2J/1053$ reproduced exactly. If confirmed in exact arithmetic, orders 1–3 give $\Delta_L \geq kd_3(\xi)$ for $g^2 \geq 3.7154$, with $d_3(1/64) = 1.902$ (recomputed here from the referee’s values rounded up). The computation of the norms is not repeated here.

The first-order loss. The true gap has no first-order term (Proposition 3.6(3)), while the lower bounds lose one because the spectral return bounds $2\Gamma(v, h')$ by $4t(v)\|Kh'\|_X$. My assessment: I think a second-order lower bound $\Delta_L \geq \kappa(3 - c\xi^2)$ is within reach of the same method, because the loss comes from one estimate that ignores the vanishing of the first-order perturbation on the plaquette eigenspace.

Clustering. If, as §H34 asserts, the coefficient of ξ^n in $\hat{C}_{pq}$ vanishes unless the face distance of p and q is at most n , then the bounds of §3 give $|\hat{C}_{pq}(\tau; \xi)| \leq (2829/13)e^{\{-13\tau/8\}(55|\xi|)^{\{2[d(p,q)/2]\}}(1 - (55|\xi|)^2)}$, uniformly in L . A written proof of that locality statement is open.

The bridges. The typed maps of §5 bring the Jacobian flow, Navier–Stokes solutions and the S^6 period data into the Hamiltonian setting. My assessment: of these, the S^6 map is the most economical, since it needs one positive number D , and the Navier–Stokes map is the most developed, since its rates are now derived conditionally (companion record). For the Jacobian map, the result that its trial states track the free two-gluon threshold identifies what the map produces; which properties of F would move the quotients outside the window $1 \pm 1/(10j)$ is Question 6.

8. Questions

1. Can the vacuum expansion of §2 be replaced by a multiscale control that persists along a path to weak coupling?
2. Can a re-centred certificate of order N pass α_N if it uses signed or same-support information?

3. Confirm the coefficient bounds of orders 3–5 by an independent, preferably exact, implementation.
4. Is there $c > 0$ with $\Delta_L \geq \kappa(3 - c\xi^2)$ for all L and all small $\xi > 0$?
5. Prove the locality statement of §H34 for the heat coefficients.
6. For the Jacobian branch, which properties of the map move the energy quotients of the trial states outside the window around the free two-gluon threshold?
7. Can a sequence of the spatial line’s kind, in which finite gaps close while spectral weight escapes to high energy, be realized along a path whose limit satisfies the reconstruction axioms of the continuum problem?

9. Verification

The scripts are in the record’s `checks/ym/` folder, with their outputs:

- `ymr_heat_constants.py` and `ymr_referee_checks.py`: the constants of Propositions 3.2 and 3.4, the tables of Proposition 3.5, the windows of Proposition 3.6, the radii family, and α_N , d_N and the thresholds of Corollary 2.6 by exact root isolation;
- `ym_order2_threshold.py`: the order-2 threshold 3.9781 with the rounding $\sqrt{3} < 1.7320509$;
- `generality_checks.py` and `workbench_reader_claims_checks.py`: Lemma 2.2 on the listed graphs (exhaustive search and linear programming), the order-2 tensor norms, and Proposition 6.1;
- the order-5 replay: the workbench’s producer (662 rows) and auditor, re-run with byte-identical output;
- `ymr_record_inventory.py`: the inventory of the Zenodo record by MD5.

References

- [A] A compact complex threefold fibred by tori over the projective line, and the six-sphere, manuscript circulated by L. Alpöge, 2026.
- [Ba] J. C. Baez, Spin network states in gauge theory, *Adv. Math.* 117 (1996) 253–272.
- [Ey] P. Eymard, L’algèbre de Fourier d’un groupe localement compact, *Bull. Soc. Math. France* 92 (1964) 181–236.
- [JW] A. Jaffe, E. Witten, Quantum Yang–Mills theory, official problem description, Clay Mathematics Institute.
- [Ka] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer, 1976.
- [KS] J. Kogut, L. Susskind, Hamiltonian formulation of Wilson’s lattice gauge theories, *Phys. Rev. D* 11 (1975) 395–408.
- [LP] G. Lumer, R. S. Phillips, Dissipative operators in a Banach space, *Pacific J. Math.* 11 (1961) 679–698.
- [O] OpenAI, Finite Time Blowup for Navier–Stokes, manuscript, 2026.
- [OS] K. Osterwalder, E. Seiler, Gauge field theories on a lattice, *Ann. Physics* 110 (1978) 440–471.
- [SZH] D. Schütte, Zheng Weihong, C. J. Hamer, Coupled cluster method in Hamiltonian lattice field theory, *Phys. Rev. D* 55 (1997) 2974.
- [Ya] D. A. Yarotsky, Quasi-particles in weak perturbations of non-interacting quantum lattice systems, *arXiv:math-ph/0411042* (2004).